MATHEMATICS · COMPUTATION · VERTEBRATE PALEONTOLOGY

ELIGULI HAN

New York University · Mathematics & Computer Science Undergraduate

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Mathematics and computer science undergraduate developing computational approaches to animal anatomy, acoustics and environmental perception. Research interests include fossil cetacean cranial anatomy, temnospondyl biomechanics, and the connection between three-dimensional geometry and biological function.

RESEARCH PUBLICATIONS

Work in progress

Current projects and working manuscripts in development, listed separately from the complete works directory.

Megalodon tooth-scale mechanics — Working technical report

Calibrated CT geometry, numerical verification and regional stress sensitivity.

Cynthiacetus digital cranial anatomy — Project in development

Complete cranial models, comparative anatomy and print-support visualisation.

Antiterrastrae biosonar simulation — Prototype in development

Interactive geometry, acoustic propagation and finite-element experiments.

Mathematics and computational models — Working manuscripts

Probability, algebra, recurrences and geometric visualisation.

Ancient Greek semantic graphs — Project in development

Inspectable language-analysis layers and corpus-based graph evaluation.

EDUCATION & MATHEMATICAL TRAINING

New York University · College of Arts & Science

Degree in progress

B.A. candidate, Mathematics & Computer Science

Completed coursework: Numerical Analysis; Partial Differential Equations; Machine Learning; Natural Language Processing.
Current coursework: Mathematical Statistics.

Awesome SummerCamp: Three consecutive participations in advanced mathematical enrichment covering geometry, algebra and number theory.

RESEARCH REPORT

Representative papers & complete mathematical notes ↓

CURRENT RESEARCH REPORT

Tooth-scale Stress Localisation in a µCT-derived Otodus megalodon Model

Revised technical report · 9 pages · 31 October 2026

SELECTED RESEARCH & COMPUTATIONAL PROJECTS

Antiterrastrae · Volumetric Biosonar & Scientific Visualization

Ongoing

Self-directed computational prototype · JavaScript, HTML/CSS, WebGL

- Developing an interactive aquatic simulation connecting three-dimensional animal surface geometry with acoustic propagation, echoes, and temperature-, salinity- and depth-dependent environmental structure.
- Implemented a tetrahedral finite-element module for scalar acoustic eigenmodes, $\mathrm{K}\mathrm{p} = \lambda \mathrm{M}\mathrm{p}$, with stiffness and consistent mass matrices, Cholesky transformation and Jacobi eigenvalue iteration.
- Integrated WebGL echo rendering with a Canvas fallback, adjustable acoustic parameters, and VTK/OBJ export of surface and volumetric geometry for inspection and further analysis.

Cynthiacetus peruvianus · Digital Cranial Anatomy

Ongoing

Independent project in development · MNHN holotype MNHN.F.PRU10

- Initiated data-access correspondence with Guillaume Billet at the Muséum national d’Histoire naturelle; availability of cranial and mandibular surface scans was confirmed, subject to a formal collections request and use conditions.
- Developing a staged research plan linking surface morphology to CT/µCT segmentation, virtual endocast reconstruction and exploratory cranial biomechanics, according to available data and documented anatomical assumptions.

Coronodon Dental Morphology · Primary-Literature Verification

2026

Independent research-verification assessment submission

- Constructed a source-verifiable question on fossil whale dental measurements; cross-checked interpretations against primary literature and documented the evidence behind a reference answer and model-error diagnosis.

TECHNICAL SKILLS & RESEARCH DIRECTION

Programming	Python, NumPy, pandas, scikit-learn; JavaScript, HTML/CSS; WebGL/Canvas; SQL; GitHub; AI-assisted development and debugging.
Methods	Numerical methods, differential equations, finite-element assembly, eigenvalue problems, three-dimensional mesh processing, scientific visualization and literature verification.
Training sought	Specimen-based CT/µCT segmentation, comparative cranial anatomy, geometric morphometrics and finite-element analysis of biological structures.

Seeking supervised undergraduate research · Summer 2027 · New York

F-1 international student